

Zetta is application-centric middleware. It is a service-oriented open source IoT platform built on Node.js combining REST API, WebSockets and reactive programming. It can run on cloud platforms like Heroku to create geo-distributed networks and stream data into machine analytics systems like Splunk.

MainFlux is application-centric middleware providing solutions based on the cloud. It has been developed as microservices, containerised by Docker and orchestrated with Kubernetes. The architecture is flexible and allows seamless integration with enterprise systems like ERP, BI and CRM. It can also integrate with databases, analytics, backend systems and cloud services easily. In addition, it supports REST, MQTT, WebSocket and CoAP.

Distributed Services Architecture (DSA) facilitates decentralised device inter-communication, allowing protocol translation and data integration to and from third party data sources.

OpenRemote is application-centric middleware used to connect any device regardless of vendor or protocol, to create meaningful connections by converting data points into smart applications. It finds use in home automation, commercial buildings, public space and healthcare. Data visualisation is integrated with devices and sensors, and turns data into smart applications.

OpenIoT is application-centric open source middleware for pulling information from sensors. It incorporates Sensing-as-a-Service for deploying and providing services in cloud environments.

ThingsBoard is an open source IoT platform for data collection, processing, data visualisation and device management. It supports IoT protocols like MQTT, CoA, and HTTP with on-premise and cloud deployment. It is horizontally scalable, stores data in Cassandra DB, HSQLDB and PostgreSQL.

NATS.io is a simple, secure and high performance open source messaging system for cloud native solutions. It is developed as microservices architecture with high performance, secure and resilient capabilities.

Benefits of open source IoT middleware

Open source middleware for the IoT has the following advantages over proprietary options:

- It is easy to upgrade to new technologies with open source middleware.
- It has the ability to connect with upcoming device protocols and backend applications.
- Open source middleware ensures lower overall software costs, and is easier to use when changing technology and open source APIs for integration.
- It has a microservices based architecture and is built using open

source technologies, resulting in high performance, scalability and fault-tolerance.

- It provides multi-protocol support and is hardware-agnostic. It supports connectivity for any device and any application.
- It has the flexibility to allow the cloud service provider to be changed.

It is very important to choose the right set of open source middleware for an IoT solution. This is a big challenge as the market offers a vast choice.

Analyse the business problem and arrive at the solution as a first step. Break the solution into services and understand the middleware needs of these services. This will help to narrow down the middleware choices.

IoT middleware helps overcome the problems associated with the heterogeneity of the entire Internet of Things by enabling smooth communication among devices and components from different vendors and based on different technologies. **END** 

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By: Dr Gopala Krishna Behara and Geeta Pradnyesh Naik

Dr Gopala Krishna Behara is a distinguished member in the Global Enterprise Architecture division of Wipro. He has a total of 22 years of IT experience.

Geeta Pradnyesh Naik is a senior enterprise architect and part of the IoT team in the Global Enterprise Architecture division of Wipro with more than 18 years of IT experience.

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